

Assignment 4 (Data Warehouse)

GitHub Project Link: <https://github.com/shreyankaseetharam/data226-assignment4-dbt>

Create a dbt project with snowflake connector (1 pt)

I first created and activated a Python virtual environment named `dbt_env`. I used a virtual environment so that the dbt packages required for this assignment would be installed separately from my other Python packages.

```
python -m pip install dbt-snowflake
```

I installed the `dbt-snowflake` package inside the activated virtual environment. This package installs dbt Core and provides the Snowflake adapter required to connect dbt to Snowflake.

I verified that dbt and the Snowflake adapter were installed correctly by running the following command:

```
dbt --version
```

The output confirmed that dbt Core version `1.12.5` and the Snowflake adapter version `1.12.1` were installed successfully.

```
(dbt_env) C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21
and Warehouse\Assignments>dbt --version
Core:
  - installed: 1.12.5
  - latest:    1.12.5 - Up to date!

Plugins:
  - snowflake: 1.12.1 - Up to date!
```

I then created a new dbt project named `build_mau` using the Snowflake connector. During the setup, I configured the project with my Snowflake account, username, database, warehouse, and role. The connection settings used were account `ntdekef-px79423`, user `SHREYANKA`, database `DEMO_DB`, warehouse `DEMO_DB`, and role `ACCOUNTADMIN`.

Creating and configuring the `build_mau` dbt project using the `dbt init` command:

```
Enter a number: 1
account (https://<this_value>.snowflakecomputing.com): ntdekef-px79423
user (dev username): SHREYANKA
[1] password
[2] keypair
[3] sso
[4] workload_identity
Desired authentication type option (enter a number): 2
private_key_path (path/to/private.key): C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\sjsu-data226-F
A26-main\keys\rsa_key.p8
private_key_passphrase (passphrase for the private key, if key is encrypted):
role (dev role): ACCOUNTADMIN
warehouse (warehouse name): DEMO_DB
database (default database that dbt will build objects in): DEMO_DB
schema (default schema that dbt will build objects in): RAW
threads (1 or more) [1]: 1
07:52:23 Profile build_mau written to C:\Users\Dell\.dbt\profiles.yml using target's profile_template.yml and your supplied values.
07:52:23 Running dbt debug to validate the project...
07:52:23 dbt version: 1.12.5
07:52:23 python version: 3.14.7
07:52:23 python path: C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\dbt_env\Scripts\python.exe
07:52:23 os info: Windows-11-10.0.26200-SP0
07:52:23 Using profiles dir at C:\Users\Dell\.dbt
07:52:23 Using profiles.yml file at C:\Users\Dell\.dbt\profiles.yml
07:52:23 Using dbt_project.yml file at C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\build_mau\dbt_
project.yml
07:52:23 adapter type: snowflake
07:52:23 adapter version: 1.12.1
07:52:23 Configuration:
07:52:23   profiles.yml file [OK found and valid]
07:52:23   dbt_project.yml file [OK found and valid]
07:52:23 Required dependencies:
07:52:23   - git [OK found]
```

After completing the dbt project configuration, I ran `dbt debug` from the root directory of the `build_mau` project. This command checked whether the project configuration, profile, required dependencies, and Snowflake connection were valid.

Successful validation of the dbt project configuration and Snowflake connection using `dbt debug`.

```

08:03:56 Connection:
08:03:56   account: ntdekef-px79423
08:03:56   user: SHREYANKA
08:03:56   database: DEMO_DB
08:03:56   warehouse: DEMO_DB
08:03:56   role: ACCOUNTADMIN
08:03:56   schema: RAW
08:03:56   authenticator: None
08:03:56   oauth_client_id: None
08:03:56   query_tag: None
08:03:56   client_session_keep_alive: False
08:03:56   host: None
08:03:56   port: None
08:03:56   proxy_host: None
08:03:56   proxy_port: None
08:03:56   protocol: None
08:03:56   connect_retries: 1
08:03:56   connect_timeout: None
08:03:56   retry_on_database_errors: False
08:03:56   retry_all: False
08:03:56   insecure_mode: False
08:03:56   reuse_connections: True
08:03:56   s3_stage_vpce_dns_name: None
08:03:56   workload_identity_provider: None
08:03:56   workload_identity_entra_resource: None
08:03:56   platform_detection_timeout_seconds: 0.0
08:03:56   Registered adapter: snowflake=1.12.1
08:04:00   Connection test: [OK connection ok]

08:04:00 All checks passed!

```

The `dbt debug` command completed successfully. The output displayed `Connection test: [OK connection ok]` and `All checks passed!`, confirming that dbt successfully connected to Snowflake using the configured settings.

Set up transform models (2pt)

I created the `models/transform` and `models/analytics` directories inside the dbt project. I then edited `dbt_project.yml` to configure how the models in each directory should be materialized.

```

models:
  build_mau:
    transform:
      +materialized: ephemeral
    analytics:
      +materialized: table

```

Input (transform) tables should be built as CTE (involves changes to dbt_project.yml)

I created `models/source.yml` to define the raw Snowflake tables that will be used as sources by the transform models.

```
version: 2

sources:
  - name: raw
    database: DEMO_DB
    schema: RAW
    description: "Schema containing the raw input tables"
    tables:
      - name: user_session_channel
        identifier: user_session_channel
        description: "Session data containing userId, session
Id, and channel"

      - name: session_timestamp
        identifier: session_timestamp
        description: "Session data containing sessionId and t
imestamp"
```

The `source.yml` file allows the SQL models to refer to the raw tables using dbt's `source()` function instead of hardcoding the complete database and schema names in every SQL file. In this project, the raw source tables are located in the `DEMO_DB.RAW` schema.

Show contents of models/transform/user_session_channel.sql and transform/session_timestamp.sql

I created `models/transform/user_session_channel.sql` with the following SQL:

```
SELECT
  userId,
  sessionId,
```

```

channel
FROM {{ source('raw', 'user_session_channel') }}
WHERE sessionId IS NOT NULL

```

This model reads the raw `user_session_channel` source table and removes records where `sessionId` is null. Because the transform directory is configured with `+materialized: ephemeral`, dbt will use this SQL as a CTE instead of creating a separate physical table.

I created `models/transform/session_timestamp.sql` with the following SQL:

```

SELECT
    sessionId,
    ts
FROM {{ source('raw', 'session_timestamp') }}
WHERE sessionId IS NOT NULL

```

This model reads the raw `session_timestamp` source table and keeps only records with a valid `sessionId`. It selects the session identifier and its timestamp for use by the final analytics model.

I validated the dbt project configuration and transform models by running: `dbt compile`

```

(dbt_env) C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\build_mau>dbt compile
09:43:54 Running with dbt=1.12.5
09:43:57 Registered adapter: snowflake=1.12.1
09:44:00 Unable to do partial parsing because saved manifest not found. Starting full parse.
09:44:04 [WARNING]: Configuration paths exist in your dbt_project.yml file which do not apply to any resources.
There are 1 unused configuration paths:
- models.build_mau.analytics
09:44:04 Found 4 models, 4 data tests, 2 sources, 562 macros
09:44:04
09:44:04 Concurrency: 1 threads (target='dev')
09:44:04

```

The command successfully parsed the dbt project and recognized the source definitions and model files. At this stage, `dbt compile` validates the SQL and dbt configuration but does not create physical tables in Snowflake.

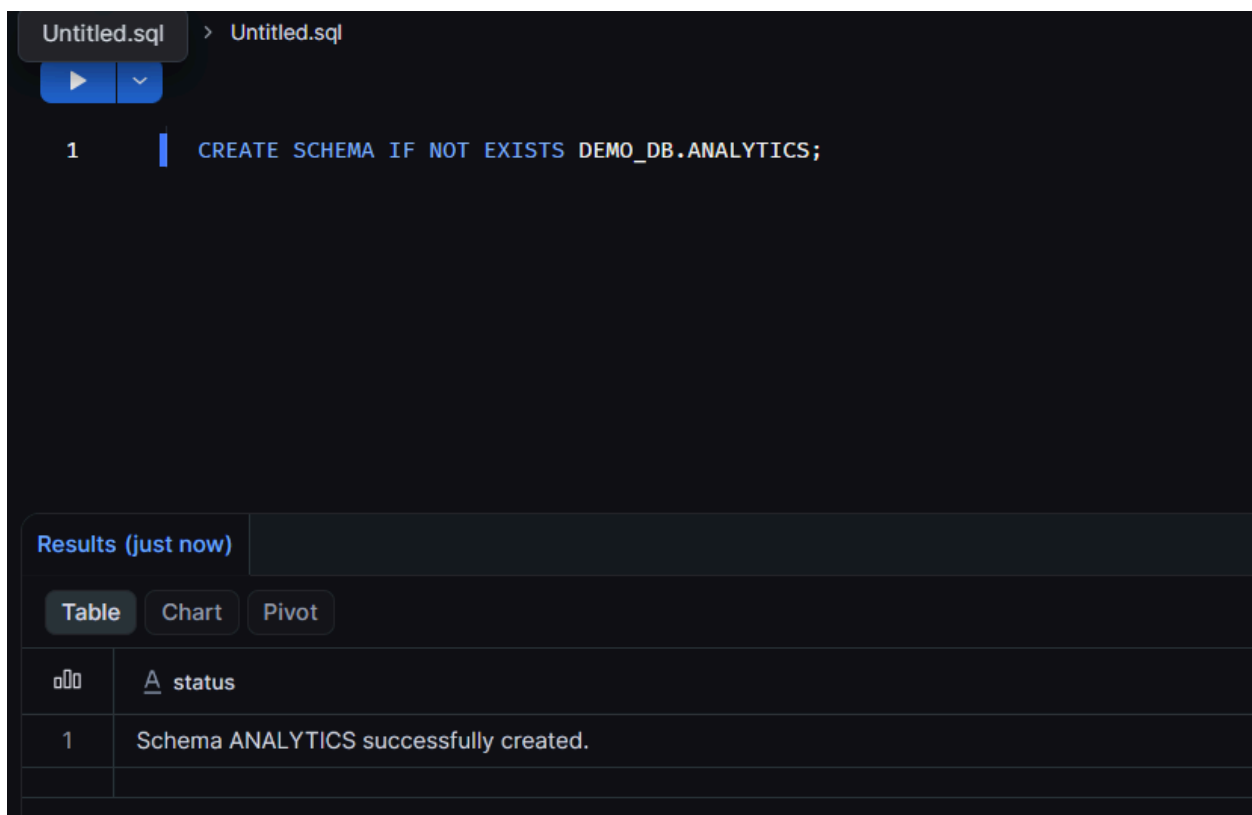
Set up analytics models (1pt)

Show contents of models/analytics/session_summary.sql

Before creating the analytics model, I verified that the `ANALYTICS` schema existed in the `DEMO_DB` database. When necessary, I created it using the following SQL statement:

```
CREATE SCHEMA IF NOT EXISTS DEMO_DB.ANALYTICS;
```

The `ANALYTICS` schema is used to store the final analytics model created by dbt. The raw input tables remain in the `RAW` schema, while the final `session_summary` table is created in the `ANALYTICS` schema.



I updated the dbt target schema from `RAW` to `ANALYTICS` so that the final analytics model would be created in the correct schema.

```

build_mau:
  outputs:
    dev:
      account: ntdekef-px79423
      database: DEMO_DB
      private_key_passphrase: sjsu
      private_key_path: C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec
21 -      Big Data and Warehouse\Assignments\sjsu-data226-FA26-main\keys
      \rsa_key.p8
      role: ACCOUNTADMIN
      schema: ANALYTICS
      threads: 1
      type: snowflake
      user: SHREYANKA
      warehouse: DEMO_DB
      target: dev

```

I created `models/analytics/session_summary.sql` to combine the cleaned session-channel data with the cleaned session-timestamp data.

The model uses dbt's `ref()` function to reference the two transform models. This allows dbt to track the dependency between the transform models and the final analytics model.

```

SELECT
  u.userId,
  u.sessionId,
  u.channel,
  st.ts
FROM {{ ref("user_session_channel") }} u
JOIN {{ ref("session_timestamp") }} st
  ON u.sessionId = st.sessionId

```

The query joins the two transform models using `sessionId`, which is the common key between them. The resulting table contains the user ID, session ID, channel, and timestamp for each session.

I ran the analytics model using: `dbt run --select session_summary`

This command builds the `session_summary` model as a physical table in the `DEMO_DB.ANALYTICS` schema. The two transform models are embedded as CTEs because they were configured as `ephemeral`.

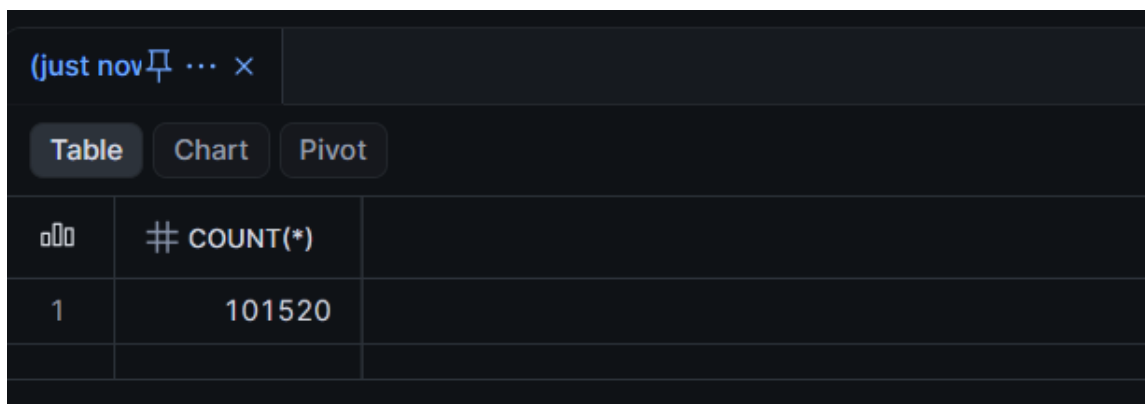
The command completed successfully and created the `session_summary` table in the `DEMO_DB.ANALYTICS` schema. The final output showed that the model passed without errors. This confirmed that the two transform models were successfully used as CTEs in the final analytics table.

```
(dbt_env) C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\bu
ild_mau>dbt run --select session_summary
14:29:45 Running with dbt=1.12.5
14:29:56 Registered adapter: snowflake=1.12.1
14:29:59 Found 3 models, 2 sources, 562 macros
14:29:59
14:29:59 Concurrency: 1 threads (target='dev')
14:29:59
14:30:03 1 of 1 START sql table model ANALYTICS.session_summary ..... [RUN]
14:30:13 1 of 1 OK created sql table model ANALYTICS.session_summary ..... [SUCCESS 101520
in 9.69s]
14:30:14
14:30:14 Finished running 1 table model in 0 hours 0 minutes and 15.71 seconds (15.71s).
14:30:14
14:30:14 Completed successfully
14:30:14
14:30:14 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=1
```

```
SELECT COUNT(*)
FROM DEMO_DB.ANALYTICS.SESSION_SUMMARY;

SELECT *
FROM DEMO_DB.ANALYTICS.SESSION_SUMMARY
LIMIT 10;
```

I verified that the analytics table was created in Snowflake by checking its row count and viewing sample records.



The screenshot shows a Snowflake query results interface. At the top, there is a search bar with the text "(just nov)". Below the search bar, there are three tabs: "Table", "Chart", and "Pivot". The "Table" tab is selected. The table has one column with the header "# COUNT(*)" and one row with the value "101520".

	# COUNT(*)
1	101520

Results (just now)

Table Chart Pivot

10 rows 1.1s

	# USERID	SESSIONID	CHANNEL	TS
	45	05ae14d7ae387b93370d142d82220f1b 0a54b19a13b6712dc04d1b49215423d8 +8 more	Facebook 10.0% Google 10.0% +3 more	4/30/2019
1	184	c41dd99a69df04044aa4e33ece9c9249	Naver	2019-05-01 00:35:59.897
2	80	fdc0eb412a84fa549afe68373d9087e9	Organic	2019-05-01 02:53:48.503
3	251	0a54b19a13b6712dc04d1b49215423d8	Facebook	2019-05-01 12:18:26.637
4	264	a914ecef9c12ffdb9bede64bb703d877	Google	2019-05-01 13:41:29.267
5	744	05ae14d7ae387b93370d142d82220f1b	Facebook	2019-05-01 14:17:54.083
6	229	eddeb82df22554fa67c641e3f8a25566	Google	2019-05-01 14:42:50.487
7	265	4c4ea5258ef3fb3fb1fc48fee9b4408c	Naver	2019-05-01 15:08:15.557
8	80	8ae1da0fe37c98412768453f82490da2	Organic	2019-05-01 15:20:27.377
9	45	60131a2a3f223dc8f4753bcc5771660c	Youtube	2019-05-01 15:53:07.017
10	198	90f4760fcc9b69c13da7368c5c2917f3	Facebook	2019-05-01 16:15:43.423

Add Snapshot to the analytics table (session_summary) (2pt)

Show contents of snapshots/snapshot_session_summary.sql

```
{% snapshot snapshot_session_summary %}
```

```
{{
  config(
    target_schema='snapshot',
    unique_key='sessionId',
    strategy='timestamp',
    updated_at='ts',
    invalidate_hard_deletes=True
  )
}}
```

```
SELECT *
FROM {{ ref('session_summary') }}
```

```
{% endsnapshot %}
```

To preserve the historical changes made to the `session_summary` analytics table, I created a dbt snapshot named `snapshot_session_summary`. The snapshot uses `sessionId` as the unique key and the `ts` column to detect changes over time. dbt automatically adds history-tracking columns such as `dbt_valid_from`, `dbt_valid_to`, `dbt_updated_at`, and `dbt_scd_id`.

```
(dbt_env) C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\bu
ild_mau>dbt snapshot
15:49:51 Running with dbt=1.12.5
15:49:57 Registered adapter: snowflake=1.12.1
15:50:01 Found 3 models, 1 snapshot, 2 sources, 562 macros
15:50:01
15:50:01 Concurrency: 1 threads (target='dev')
15:50:01
15:50:09 1 of 1 START snapshot snapshot.snapshot_session_summary ..... [RUN]
15:50:15 1 of 1 OK snapshot snapshot.snapshot_session_summary ..... [SUCCESS 101520
in 5.70s]
15:50:16
15:50:16 Finished running 1 snapshot in 0 hours 0 minutes and 15.58 seconds (15.58s).
15:50:16
15:50:16 Completed successfully
15:50:16
15:50:16 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=1
```

I executed the snapshot using the `dbt snapshot` command. The snapshot completed successfully and created the `snapshot.snapshot_session_summary` table. The output showed `PASS=1`, `ERROR=0`, and `Completed successfully`, confirming that the snapshot was configured correctly.

```
SELECT *
FROM DEMO_DB.SNAPSHOT.SNAPSHOT_SESSION_SUMMARY
LIMIT 10;
```

Results (just now)

Table Chart Pivot

10 rows 1.2s

	# USERID	SESSIONID	CHANNEL	TS	DBT_SCD_ID	DBT_UPDATED_AT	DBT_VALID_FROM	DBT_VALID_TO
	45	744						
1	184	c41dd99a69df04044aa4e33ece9c9249	Naver	2019-05-01 00:35:59.897	cbf977c5099c3b5feb71ec791dbfb857	2019-05-01 00:35:59.897	2019-05-01 00:35:59.897	null
2	80	fdc0eb412a84fa549afe68373d9087e9	Organic	2019-05-01 02:53:48.503	f1106b98f87644982e9b5e82f5454ea1	2019-05-01 02:53:48.503	2019-05-01 02:53:48.503	null
3	251	0a54b19a13b6712dc04d1b49215423d8	Facebook	2019-05-01 12:18:26.637	b4706917a7e8ee94cc25608c9bbfcbf9	2019-05-01 12:18:26.637	2019-05-01 12:18:26.637	null
4	264	a914ecef9c12fddb9bede64bb703d877	Google	2019-05-01 13:41:29.267	61ffaf490c97e7e01d3213f0e8a0dae6	2019-05-01 13:41:29.267	2019-05-01 13:41:29.267	null
5	744	05ae14d7ae387b93370d142d82220f1b	Facebook	2019-05-01 14:17:54.083	5a2be3fa45dbf6a9bd44b754ed6b04b7	2019-05-01 14:17:54.083	2019-05-01 14:17:54.083	null
6	229	eddeb82df22554fa67c641e3f8a25566	Google	2019-05-01 14:42:50.487	b567b92113cb4455c43df2dc3e405a09	2019-05-01 14:42:50.487	2019-05-01 14:42:50.487	null
7	265	4c4ea5258ef3fb3fb1fc48fee9b4408c	Naver	2019-05-01 15:08:15.557	ad590dd472ea36c4d734a0acabee74549	2019-05-01 15:08:15.557	2019-05-01 15:08:15.557	null
8	80	8ae1da0fe37c98412768453f82490da2	Organic	2019-05-01 15:20:27.377	38aa4b4921dd2bdf75c8863d036d2f21	2019-05-01 15:20:27.377	2019-05-01 15:20:27.377	null
9	45	60131a2a3f223dc8f4753bcc5771660c	Youtube	2019-05-01 15:53:07.017	291298ba2569a2de36837f6010ea91fc	2019-05-01 15:53:07.017	2019-05-01 15:53:07.017	null

I verified the snapshot table in Snowflake by querying `DEMO_DB.SNAPSHOT.SNAPSHOT_SESSION_SUMMARY`. The results displayed the original session fields along with dbt's history-tracking fields, including `DBT_SCD_ID`, `DBT_UPDATED_AT`, `DBT_VALID_FROM`, and `DBT_VALID_TO`. This confirmed that the snapshot was created successfully and is preserving the historical state of the `session_summary` model.

Add at least 2 tests to sessionId field of session_summary (2pt)

Show contents of models/schema.yml

```
version: 2

models:
  - name: session_summary
    description: "Analytics model for session data"
    columns:
      - name: sessionId
        description: "Unique identifier for each session"
        data_tests:
          - unique
          - not_null
```

I created `models/schema.yml` to define data-quality tests for the `sessionId` field in the `session_summary` model. I added a `unique` test to verify that each session identifier

appears only once and a `not_null` test to verify that no session identifier is missing.

```
(dbt_env) C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\build_mau>dbt run
16:36:18 Running with dbt=1.12.5
16:36:20 Registered adapter: snowflake=1.12.1
16:36:23 Found 3 models, 1 snapshot, 2 data tests, 2 sources, 562 macros
16:36:23 Concurrency: 1 threads (target='dev')
16:36:23
16:36:29 1 of 1 START sql table model ANALYTICS.session_summary ..... [RUN]
16:36:34 1 of 1 OK created sql table model ANALYTICS.session_summary ..... [SUCCESS 101520
in 4.55s]
16:36:36
16:36:36 Finished running 1 table model in 0 hours 0 minutes and 13.07 seconds (13.07s).
16:36:36
16:36:36 Completed successfully
16:36:36
16:36:36 Done. PASS=1 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=1
```

```
(dbt_env) C:\Users\Dell\Desktop\San Jose Fall 2026\DATA-226 Sec 21 - Big Data and Warehouse\Assignments\build_mau>dbt test
16:37:07 Running with dbt=1.12.5
16:37:09 Registered adapter: snowflake=1.12.1
16:37:10 Found 3 models, 1 snapshot, 2 data tests, 2 sources, 562 macros
16:37:10 Concurrency: 1 threads (target='dev')
16:37:10
16:37:13 1 of 2 START test not_null_session_summary_sessionId ..... [RUN]
16:37:15 1 of 2 PASS not_null_session_summary_sessionId ..... [PASS in 1.73s]
16:37:15 2 of 2 START test unique_session_summary_sessionId ..... [RUN]
16:37:15 2 of 2 PASS unique_session_summary_sessionId ..... [PASS in 0.61s]
16:37:16
16:37:16 Finished running 2 data tests in 0 hours 0 minutes and 6.47 seconds (6.47s).
16:37:17
16:37:17 Completed successfully
16:37:17
16:37:17 Done. PASS=2 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=2
```

I ran `dbt test` to execute the data-quality tests. Both tests passed successfully, confirming that the `sessionId` values in `session_summary` are unique and contain no null values.